GLUCOSE AND INSULIN: HOW A SPIKY RELATIONSHIP SMOOTHED OUT

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Abstract. Many Americans live with Type 1 diabetes, a condition characterized by the body's inability to produce insulin. Effective management requires precise glucose monitoring and insulin delivery to avoid life-threatening complications. In this project, we design and simulate an automated "artificial pancreas" using a system of linear ordinary differential equations to model blood sugar dynamics. We implement a Linear Quadratic Regulator (LQR) to optimize insulin delivery by solving the infinite time horizon algebraic Riccati equation at each time step. To account for different physiological states, we utilize two distinct control regimes - maintenance and hyperglycemic - switching costs and targets based on current glucose levels. Real life continuous glucose monitors have a 30-minute sensor lag to the actual levels of glucose in the blood. We solve this issue of the 30-minute sensor lag by integrating a Kalman filter to provide real-time state estimation. Our results demonstrate that this combined approach effectively eliminates insulin oscillations caused by the time delays and maintains tighter glycemic control than simpler models. Comparison with real-world patient data helps validate our model's stability and its potential for improving automated glucose regulation in the blood stream.

1. Background

Approximately 2 million Americans suffer from type 1 diabetes[6] and are unable to produce insulin which is critical for glucose metabolism. If left untreated, high blood sugar can cause serious complications including blindness and death. In order to stay healthy and functional, diabetics must measure their blood sugar and inject insulin when needed. Sub-dermal Continuous Glucose Monitors (CGMs) monitor the interstitial glucose concentration, replacing the need for frequent finger pricks. More advanced machines, including a hybrid loop system or an artificial pancreas, are able to both monitor blood sugar and dispense insulin depending on the user's needs[2].

In this project we simulate an artificial pancreas with the goal of optimizing automated insulin injection using Linear Quadratic Regulator (LQR), the algebraic Ricatti equation, and the Kalman filter.

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2. Mathematical Representation

2.1. **Blood sugar model.** In order to model an individual's blood sugar we use the following system of linear ordinary differential equations:

$$\dot{x_1} = -ax_1 - bx_2$$
$$\dot{x_2} = dx_1 - cx_2 + u$$

Var.	**Physical Meaning**	**Units**
x_1	Glucose	mg/dL
x_2	Insulin	μU/mL
$\dot{x}_1$	Glucose Rate of Change	mg/dL/min
$\dot{x}_2$	Insulin Rate of Change	μU/mL/min
a	Insulin-Indep. Clearance	1/min
b	Insulin Sensitivity	(mg/dL/min)/(μU/mL)
c	Insulin Degradation Rate	1/min
d	Endogenous Production	(μU/mL/min)/(mg/dL)
u	Insulin Input (Control)	μU/mL/min

Table 1. Model variables and units

2.2. **Glucose spikes.** To simulate a person's glucose levels throughout the day, we developed a function to create a randomized "day" of glucose spikes. The first spike represents the dawn phenomenon, after which positive spikes signify meals and negative spikes signify exercise. This function is later used as a proxy for a human being who eats three meals and a snack a day and exercises once a day. The values for spikes were drawn from medical literature$_{8,3}$ and our diabetic team member's empirical observations.

Event	**Time distribution**	**Spike**	**Scale distribution**	**Duration distribution**
Dawn Phenomenon	$U(3,8)$	20	$U(0,2)$	NA
Breakfast	$U(7,9)$	40	$U(0,2)$	$U(6,12)$
Lunch	$U(11,13)$	60	$U(0,2)$	$U(6,24)$
Snack	$U(6,23)$	20	$U(0,2)$	$U(6,9)$
Exercise	$U(6,23)$	-35	$U(0,2)$	$U(1,12)$
Dinner	$U(18,20)$	80	$U(0,2)$	$U(6,36)$

Table 2. Glucose spike simulation

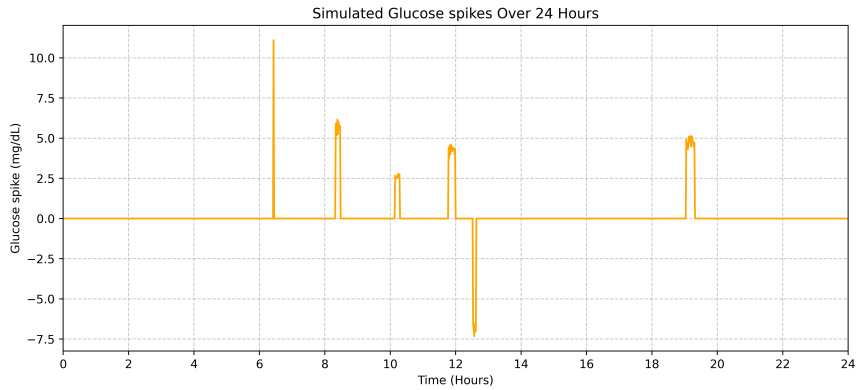

FIGURE 1. Simulated Glucose spikes

2.3. **No control check.** In order to verify that our equations and glucose spike simulation are reasonable physically, we simulate a 24 hour period without any control.

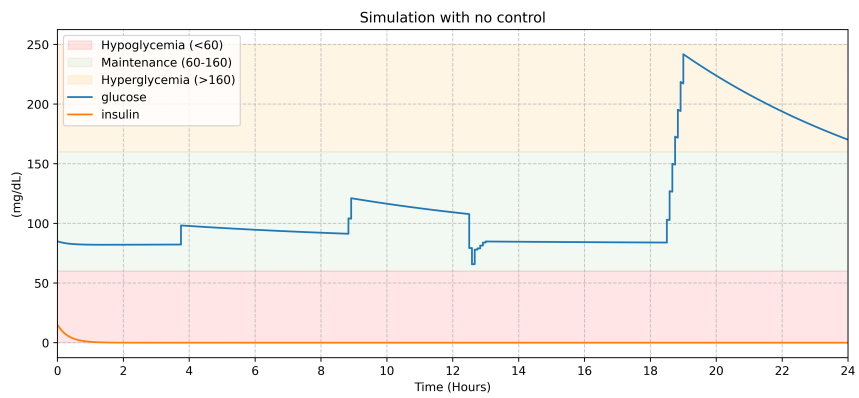

FIGURE 2. Simulation with no control

Despite the insulin concentration going to zero, blood glucose concentration will naturally decrease slowly as the body uses it for fuel. Note that this rate is entirely dependent on the parameter a, and that the insulin concentration goes to 0 since there is no endogenous insulin production; represented by $d = 0$ in the model.

3. SOLUTION

3.1. **Assumptions.** We assume the following to make the problem tractable:

- The artificial pancreas can dispense insulin$_1$.
- Blood glucose levels are measured every five minutes$_2$ with a 30 minute lag$_3$.
- Initial glucose and insulin levels are known.
- The artificial pancreas internally converts the concentration u into units of volume.

Many of these assumptions are coded as parameters in our simulation class and could be changed easily to reflect the user's experience. For example, if an artificial pancreas takes measurements every 10 minutes, this could simply be accounted for in the model by changing the corresponding parameters.

3.2. **Discretization.** Since our ODEs are continuous and the artificial pancreas takes discrete measurements, we used `scipy.signal` to discretize our ODEs. This lead to the following state equation and matrices:

$$x[k+1] = A_d x[k] + B_d u[k],$$
$$A_d = e^{(A \cdot dt)} \text{ and } B_d = \left(\int_0^{\Delta t} e^{A\tau} d\tau \right) B.$$

These equations represent the continuous state evolution between each measurement as an integral and allow us to represent our continuous system in a discrete setting.

3.3. **Optimization Problem.** Ultimately, we wish to minimize the amount of insulin the artificial pancreas uses while maintaining a healthy glucose level in the bloodstream. Thus, we want to penalize differences between ideal levels of glucose and insulin with the observed levels, and this leads to the following functional:

$$\int_0^{t_f} [\mathbf{x}^T Q \mathbf{x} + \mathbf{u}^T R \mathbf{u}] dt$$

where $Q = \begin{bmatrix} g & 0 \\ 0 & i \end{bmatrix}$ and $R = \begin{bmatrix} c_1 & 0 \\ 0 & c_2 \end{bmatrix}$. The constants g, i are the costs imposed on the differences in observed glucose and insulin levels from the target levels of each; the constants c_1, c_2 are the controllers on the amounts of insulin and glucose injected, respectively. This setup allows us to minimize our functional with LQR, and a plot of the results is shown below.

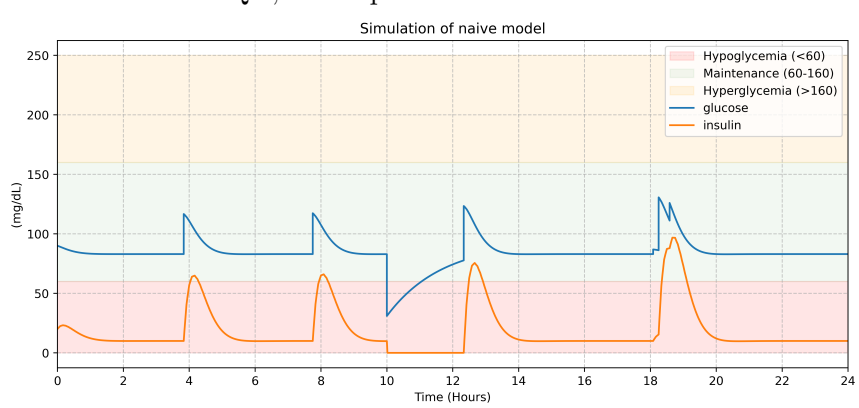

FIGURE 3. Insulin control

We have set up our model to solve the algebraic Ricatti equation every five minutes, corresponding to solving the infinite time horizon problem every

five minutes. We do this to update the optimal control; changes in glucose will change the amount of insulin needed. If there were no more unexpected changes in glucose levels such as meals or snacks we would only need to solve the equation once. With a single solution or insulin injection them glucose levels would stabilize. However, due to meals, exercise, and other external factors we need to update the optimal control as well. This results in solving the infinite time horizon every five minutes, which is the frequency that blood glucose levels are measured.

3.4. **Time Delays.** Our model's performance suffered after time delays were introduced. It overcompensated which led to oscillations in insulin levels.

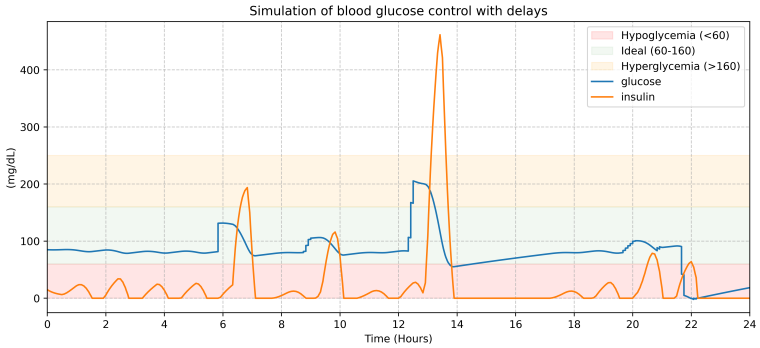

FIGURE 4. Simulation with 30 minute time delay

We found that by drastically increasing the controller penalty we could eliminate the insulin oscillations, but this amounted to applying almost no control.

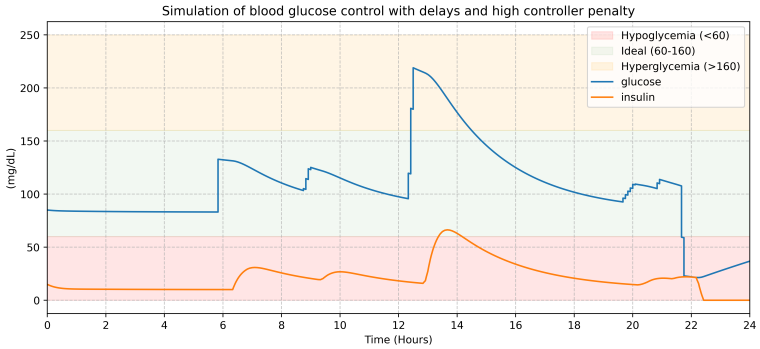

FIGURE 5. Simulation with 30 minute time delay and high controller penalty

3.5. **Kalman Filter.** In order to combat the oscillations, we added a Kalman filter which takes in the delayed glucose reading and predicts its evolution in time. This eliminated the oscillations while still allowing for control. Our

implementation of the Kalman filter came from the LQG_{10} lab and follows this algorithm:

$$
\begin{aligned}
\textbf{Fit:} \qquad L_k &= S_k H^T (H S_k H^T + D)^{-1} \\
S_{update} &= (I - L_k H) S_k \\
S_{k+1} &= A S_{update} A^T + W \\
\textbf{Predict:} \qquad \hat{x}_{k|k-1} &= A\hat{x}_{k-1} + Bu_{k-1} \\
\textbf{Update:} \qquad \hat{x}_k &= \hat{x}_{k|k-1} + L_k(z_k - H\hat{x}_{k|k-1})
\end{aligned}
$$

The fit equation is pre-computable given the desired number of time steps and so the method is well-designed for fringe deployment.

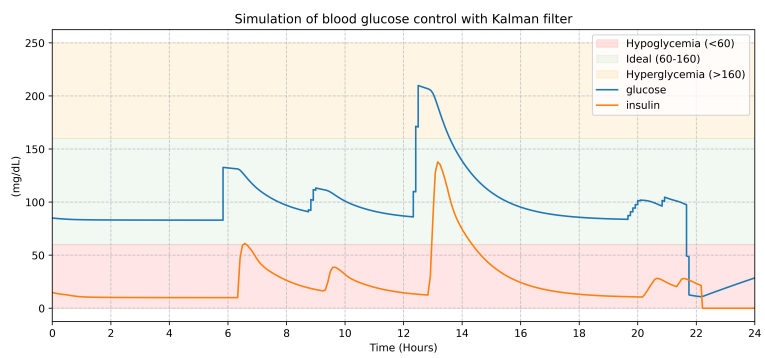

FIGURE 6. Simulation with 30 minute time delay and Kalman Filter

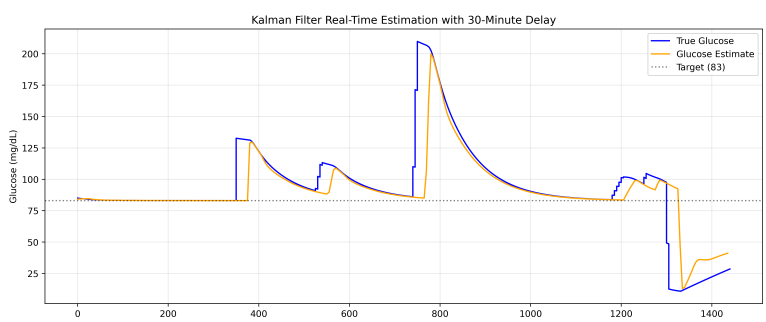

FIGURE 7. Kalman Filter's predictions compared with true concentrations

While not exact, our Kalman filter accurately predicts how spikes will evolve with only a slight lag. This lag is due to the fact that the CGM only takes a measurement every five minutes, and so the Kalman filter does not update until a maximum of five minutes after the spike has begun.

3.6. **Control Regimes.** Not all deviations from the desired blood glucose level are created equal; depending on the concentration of glucose in blood we have two separate gain matrices to calculate the optimal control. We treat glucose levels levels between 60 and 160 as maintenance, and above 160 are considered hyperglycemic. When user's have low blood sugar levels such as below 60, they should eat a high carbohydrate snack to regulate blood sugar. Because it is physically exhausting when blood sugar levels change rapidly, when an individual is near the desired level, it is preferable to have gradual change, as it is more comfortable. Yet when blood sugar levels are too high, safety must be prioritized before comfort. Thus we do not penalize the rate of glucose change when levels are hyperglycemic, as a quick change is preferable to potential death or injury.

Regime	**Range**	**Control / Cost**
Hyperglycemic	> 160	**Insulin:** Distance from target state
Maintenance	$60 - 160$	**Insulin:** Distance from target state and derivative of glucose level

TABLE 3. Control regimes

For the maintenance model we augmented the states to include the glucose state from the previous time step:

$$x_{k,aug} = \begin{bmatrix} G_k - G_{target} \\ I_k - I_{base} \\ G_{k-1} - G_k \end{bmatrix}$$

$$A_2 = \begin{bmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ 1 & 0 & 0 \end{bmatrix} \; B_2 = \begin{bmatrix} b_{11} \\ b_{21} \\ 0 \end{bmatrix} \; Q_2 = \begin{bmatrix} q_G + q_D & 0 & -q_D \\ 0 & q_I & 0 \\ -q_D & 0 & q_D \end{bmatrix}$$

With q_G being the glucose penalty, q_D the derivative penalty and q_I the insulin penalty. The final optimal control is given by $u_k = -K_2 x_{k,aug} + I_{base}$. Since the gain matrices are pre-computable, the regime model is not appreciably more computationally expensive.

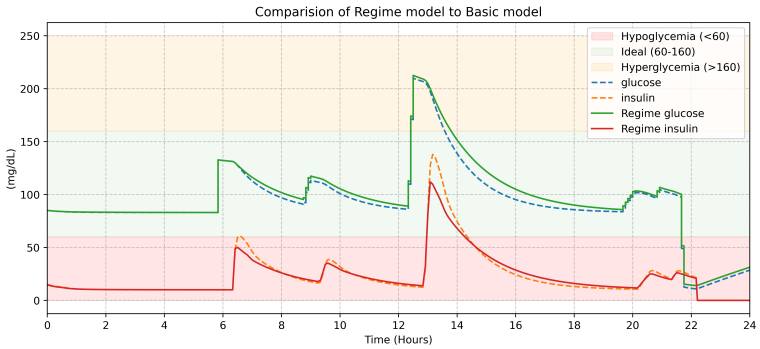

FIGURE 8. Comparison of regime and basic models

Comparing the updated model to the basic model we see that the updated model requires less insulin and has a milder slope. This increases user comfort and saves money.

4. Interpretation

4.1. **Comparison to real world data.**

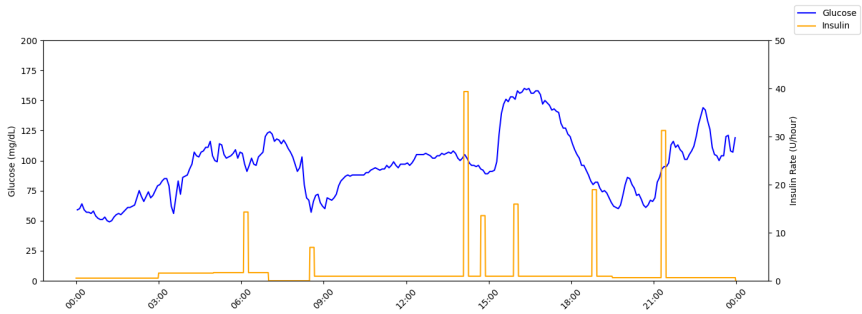

FIGURE 9. Blood glucose and insulin injection data for Kurt Healy on March 30, 2026

A day of real-world glucose and insulin data for ACME student, Kurt Healy, who is a type 1 diabetic. Kurt uses a basal-bolus approach, where he has an automatic small drip of insulin every 5 minutes and then manually injects higher levels of insulin when he eats a meal, or otherwise gets out of his desired range. We see that our model provides tighter blood sugar control than the standard basal-bolus approach that type 1 diabetics use since it acts much quicker to alleviate spikes. However, our model has the disadvantage that Kurt is able to anticipate when he will spike and add extra insulin beforehand, while our model will play catch up once the first insulin spike is registered.

Verifying our model provides tighter control could be done in two ways. We could implement a simulation that only used a small preprogrammed drip and large insulin boluses as needed. We could also obtain data from

a diabetic that uses an artificial pancreas system similar to the one we modeled.

A basal-bolus simulation was not implemented as it is nonlinear and wouldn't allow us to solve the algebraic Ricatti equation, falling outside the scope of this project. Kurt's CGM does not have the capability to communicate to his pump and thus cannot release insulin which is the control in our model. Directly comparing our implementation to the real-world data is infeasible, as we simulated a system of ODEs with control and the real-world data is the results of an unknown system of ODEs with a different control.

4.2. **Conclusion.** The equations we used for our model were a simplified version of the Non-Linear Bergman Minimal equations which describe two types of insulin, allowing for the delayed effect of insulin.

$$\dot{G} = -(p_1 + X)G + p_1 G_b$$
$$\dot{X} = -p_2 X + p_3(I - I_b) \text{ and } \dot{I} = -n(I - I_b) + u$$

where G is the blood glucose, I is the blood insulin and X is the active insulin. Real world systems use Model Predictive Control (MPC) algorithms solve ODE's in real time and calculate optimal control "in the loop"[5]. Further study could include implementing clinical trials with our model and seeing how the results compare to the standard approach. Another area of future work would be to include an artificial pancreas that can dispense both insulin and glucose, allowing for another control of adding glucose to the blood stream when levels are too low. Implementing these methods are the next steps in progression towards a more realistic controller.

In conclusion, the integration of Linear Quadratic Regulators and Kalman filtering provides a computationally efficient and stable framework for automated glucose regulation. By implementing a multi-regime control strategy, our model balances the need for aggressive intervention during hyperglycemic phases with metabolic "smoothing" during maintenance phases; resulting in improved performance over a single regime model.

Our findings highlight that while time delays inherent to CGM sensors can introduce dangerous oscillations in a naive control loop, the Kalman filter effectively mitigates these delays by estimating states in real-time. While our model is linear and cannot capture all of the non-linear dynamics of human metabolism, it demonstrates that a hybrid control-estimation approach still maintains tight glycemic control. This project underscores the potential for control theory to bridge the gap between reactive diabetes management and proactive physiological stabilization.

References

(1) https://pmc.ncbi.nlm.nih.gov/articles/PMC5467105/
(2) https://pmc.ncbi.nlm.nih.gov/articles/PMC6501529/
(3) https://pmc.ncbi.nlm.nih.gov/articles/PMC3005050/
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(10) https://labs.acme.byu.edu/Volume4/LQG/LQG.html

5. Code

All of our code is available at:
https://github.com/Jaredwinn77/ACME_VOL4_Diabetes/tree/main